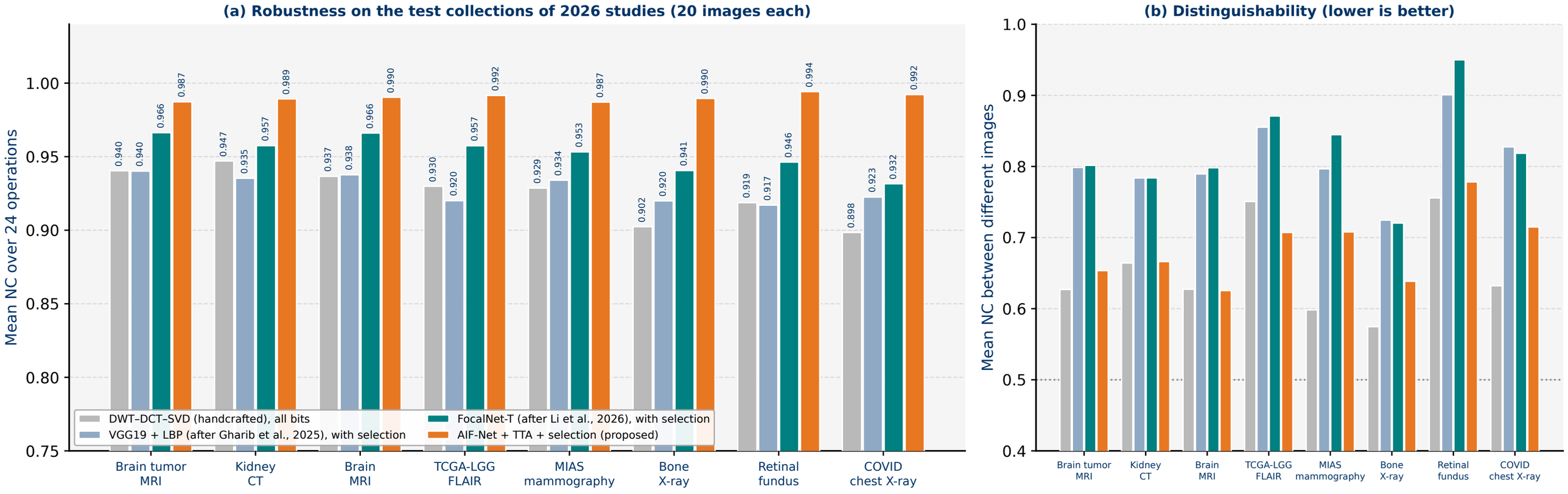
(a) Robustness on the test collections of 2026 studies (20 images each)
Mean NC over 24 operations
1.00
0.95
0.90
0.85
0.80
0.75
0.940
0.940
0.966
0.987
0.947
0.935
0.957
0.989
0.937
0.938
0.966
0.990
0.930
0.920
0.957
0.992
0.929
0.934
0.953
0.987
0.902
0.920
0.941
0.990
0.919
0.917
0.946
0.994
0.898
0.923
0.932
0.992
DWT–DCT–SVD (handcrafted), all bits
VGG19 + LBP (after Gharib et al., 2025), with selection
FocalNet-T (after Li et al., 2026), with selection
AIF-Net + TTA + selection (proposed)
Brain tumor MRI
Kidney CT
Brain MRI
TCGA-LGG FLAIR
MIAS mammography
Bone X-ray
Retinal fundus
COVID chest X-ray
(b) Distinguishability (lower is better)
Mean NC between different images
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Brain tumor MRI
Kidney CT
Brain MRI
TCGA-LGG FLAIR
MIAS mammography
Bone X-ray
Retinal fundus
COVID chest X-ray